E-Commerce Business Insights

SQL Project

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1. Executive Summary

This project demonstrates end-to-end SQL-based analysis of a simulated e-commerce database. Using advanced querying techniques—including window functions, CTEs, cohort analysis, and RFM segmentation—the analysis surfaces actionable business insights across five key dimensions.

Key Performance Highlights

Metric	Value	Trend
Total Customers	1,247	+18% YoY
Annual Revenue	$1.05M	+31% YoY
Avg Order Value	$142.80	+7% YoY
Repeat Customer Rate	57%	+4 pts YoY
Top Region	East	341 customers
Best-Selling Product	Wireless Earbuds	2,341 units

Summary of Findings

- The East region has the highest customer concentration (27%), driven by strong urban adoption.
- Revenue shows consistent month-over-month growth, with Q4 accounting for 38% of annual revenue.
- Electronics is the top-performing category, contributing 64% of total revenue.
- Repeat customers generate 3.2x the lifetime value of new customers.
- RFM segmentation identified 213 'Champion' customers who warrant priority retention programmes.

2. Project Overview

2.1 Objectives

The project was designed to answer five critical business questions using structured data analysis:

1. Understand how customers are distributed across geographic regions.
2. Track and forecast monthly revenue trends over time.
3. Identify best-selling products and top-performing categories.
4. Calculate Customer Lifetime Value (CLV) and segment customers accordingly.
5. Differentiate between new and repeat customers to assess retention health.

2.2 Database Schema

The analysis operates on four normalised relational tables:

Table	Primary Key	Key Columns
Customers	CustomerID	Name, Region, JoinDate
Orders	OrderID	CustomerID, OrderDate, TotalAmount
OrderItems	ItemID	OrderID, ProductID, Quantity, Price
Products	ProductID	ProductName, Category

2.3 Tools & Technologies

- Database: MySQL 8.0+ (compatible with PostgreSQL with minor syntax adjustments)
- SQL Features: CTEs, Window Functions (RANK, LAG, NTILE, FIRST_VALUE), Subqueries, Aggregations
- Visualisation: Microsoft Excel / Tableau (charts referenced in dashboard deliverable)
- Documentation: This report (Microsoft Word)

3. Analysis Methodology

3.1 Customer Distribution by Region

Regional analysis uses COUNT aggregation with a window-function denominator to compute percentage share without a subquery. The query surfaces both absolute counts and relative market share per region.

SQL Technique: GROUP BY + Window Function (SUM OVER()) for percentage calculation.

3.2 Monthly Revenue Trends

Revenue trends are computed using DATE_FORMAT to bucket orders by month. Month-over-month growth is derived using the LAG window function, which accesses the prior row's value without a self-join. A rolling cumulative YTD figure is computed using SUM OVER with ORDER BY clause.

SQL Technique: DATE_FORMAT bucketing, LAG() for MoM delta, running SUM() for YTD.

3.3 Best-Selling Products

Product ranking combines JOIN across OrderItems and Products tables, with SUM(Quantity) as the ranking metric. RANK() OVER enables competitive ranking within and across categories, identifying both absolute leaders and per-category champions.

SQL Technique: Multi-table JOIN, SUM aggregation, RANK() OVER PARTITION BY.

3.4 Customer Lifetime Value (CLV)

CLV is computed as the sum of all TotalAmount values per customer. A CTE-based approach then applies PERCENTILE_CONT to define segment thresholds dynamically, classifying customers into Bronze, Silver, Gold, and Platinum tiers based on their spending quantile.

SQL Technique: CTEs, PERCENTILE_CONT, CASE-WHEN segmentation logic.

3.5 New vs. Repeat Customer Analysis

Customers are classified by order count. A cohort analysis CTE tracks each customer's first order date, then compares it to subsequent order dates to count new vs. repeat purchasers per month. Retention rate is calculated as a percentage of each monthly cohort that returned in subsequent periods.

SQL Technique: MIN() to define cohort, conditional SUM() for classification, FIRST_VALUE() for retention baseline.

4. Results & Insights

4.1 Customer Distribution by Region

Region	Customers	% Share	Trend
East	341	27.3%	+22% YoY
North	312	25.0%	+15% YoY
West	305	24.5%	+19% YoY
South	289	23.2%	+16% YoY

The East region leads by a narrow margin. All regions show strong double-digit YoY growth, suggesting broad market expansion rather than concentration risk.

4.2 Monthly Revenue Trend (2024)

Month	Revenue	MoM Growth	Cumulative YTD
Jan	$45,200	—	$45,200
Feb	$52,100	+15.3%	$97,300
Mar	$61,400	+17.9%	$158,700
Apr	$58,900	−4.1%	$217,600
May	$74,300	+26.1%	$291,900
Jun	$83,700	+12.7%	$375,600
Jul	$91,200	+9.0%	$466,800
Aug	$88,400	−3.1%	$555,200
Sep	$96,100	+8.7%	$651,300
Oct	$107,500	+11.9%	$758,800
Nov	$132,200	+23.0%	$891,000
Dec	$158,600	+20.0%	$1,049,600

Revenue grew steadily throughout the year, with a seasonal dip in April and August. Q4 (Oct–Dec) represented 38% of annual revenue, confirming strong holiday-season demand.

4.3 Best-Selling Products

Rank	Product	Category	Units Sold	Revenue
1	Wireless Earbuds	Electronics	2,341	$117,050
2	Smart Watch	Electronics	1,876	$375,200
3	Laptop Stand	Accessories	1,654	$82,700
4	USB Hub	Accessories	1,432	$57,280
5	Phone Case	Accessories	1,289	$38,670
6	Mechanical Keyboard	Electronics	987	$197,400

4.4 Customer Lifetime Value

- Average CLV across all customers: $847
- Platinum segment (top 10%): avg CLV of $3,200+
- East region commands the highest average CLV at $912
- Customers with 3+ orders have a CLV 4.7x that of one-time buyers

4.5 New vs. Repeat Customers

Segment	Customers	% of Total	Avg CLV
New (1 order)	534	42.8%	$210
Returning (2–4 orders)	561	45.0%	$720
Loyal (5+ orders)	152	12.2%	$2,850

5. Business Recommendations

Based on the data analysis, the following strategic actions are recommended:

Retention & Loyalty

6. Launch a tiered loyalty programme targeting the 'Returning' segment (45% of customers) to convert them to 'Loyal' status—this could increase average CLV by an estimated $2,130 per customer.
7. Deploy a win-back email campaign for the 389 customers at risk (90+ days inactive), prioritising those with high historical spend.

Product Strategy

8. Increase inventory and marketing budget for Wireless Earbuds and Smart Watch—both show high volume and strong revenue contribution.
9. Bundle Laptop Stand and USB Hub (frequently bought together) to increase average order value.

Regional Expansion

10. South region lags by ~15% in customer count despite similar market size; targeted regional campaigns could close the gap.
11. East region's success should be studied and replicated—identify which acquisition channels drive its superior CLV.

Seasonal Planning

12. Prepare inventory and staffing 6 weeks before November to capitalise on Q4 demand (38% of annual revenue).
13. Use April and August lulls for clearance promotions to smooth revenue curve.

6. Technical Appendix

6.1 Advanced SQL Techniques Used

Technique	Application
CTEs (WITH clause)	Modular query structure for CLV segmentation and cohort analysis
Window Functions	RANK(), LAG(), NTILE(), FIRST_VALUE() for ranking and trend analysis
Cohort Analysis	Customer retention tracking using MIN(OrderDate) as cohort anchor
RFM Segmentation	NTILE(5) scoring on Recency, Frequency, Monetary axes
Product Affinity	Self-JOIN on OrderItems to find co-purchased products
PERCENTILE_CONT	Dynamic CLV segment thresholds based on data distribution

6.2 Script File Reference

The complete SQL script (ecommerce_analysis.sql) is structured in six sections:

- Section 0 — Database setup and sample data insertion
- Section 1 — Customer distribution queries
- Section 2 — Monthly revenue trend analysis
- Section 3 — Best-selling product queries
- Section 4 — Customer Lifetime Value analysis
- Section 5 — New vs. repeat customer classification
- Section 6 — Bonus: RFM analysis, product affinity, revenue-at-risk

6.3 Deliverables Checklist

Deliverable	Status
SQL Analysis Script (ecommerce_analysis.sql)	Complete
Project Report (this document)	Complete
Interactive HTML Dashboard	Complete
Database Schema Diagram	Included in Section 2.2
Business Recommendations	Included in Section 5